The Role of Dealer-Like Liquidity Providers in Absorbing Temporary Liquidity Shocks

Project Proposal

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1 Research Question

Do dealer-like liquidity providers absorb temporary liquidity shocks, and does their liquidity provision contribute to the subsequent stabilization of prices and market liquidity?

The project investigates whether observable liquidity-provision behaviour in a limit order book is consistent with the traditional role of dealers as liquidity providers. In particular, we study whether liquidity providers respond contrarianly to temporary order-flow pressure and whether such behaviour is associated with subsequent recovery in prices and market liquidity.

2 Motivation

Dealers and other liquidity providers facilitate trading by standing ready to take the opposite side of market orders. However, liquidity provision exposes them to inventory risk and adverse selection. Consequently, liquidity provision may become particularly important during temporary episodes of extreme buying or selling pressure.

The project combines the dealer-market perspective of Ho and Stoll [1] and Amihud and Mendelson [2] with the order-book dynamics of Biais and Weill [3] and the empirical evidence on liquidity suppliers in Biais, Declerck and Moinas [4].

Conceptually, the project studies the following mechanism:

Liquidity Shock $\longrightarrow$ Liquidity Provision $\longrightarrow$ Shock Absorption $\longrightarrow$ Price and Liquidity Stabilization.

3 Objectives

The main objectives are:

1. Construct measures of market liquidity and order-flow pressure from high-frequency limit order book data.
2. Identify temporary liquidity shocks using extreme order-flow imbalance and related market conditions.
3. Examine the behaviour of liquidity providers following such shocks, with particular emphasis on contrarian liquidity provision.

4. Measure the subsequent response of prices, bid–ask spreads and order-book depth.

5. Investigate whether stronger liquidity provision is associated with faster stabilization of prices and market liquidity.

6. Examine the adverse-selection exposure associated with liquidity provision where order-level data permit.

4 Hypotheses

H1: Liquidity deterioration
Temporary liquidity shocks are associated with wider bid–ask spreads, lower order-book depth and stronger order-flow imbalance.

H2: Contrarian liquidity provision
Following an extreme buying or selling shock, liquidity providers exhibit behaviour in the opposite direction to the prevailing order-flow pressure.

H3: Market stabilization
Greater liquidity provision following a liquidity shock is associated with faster subsequent stabilization of prices and market liquidity.

H4: Adverse selection
Liquidity provision during severe market imbalance is associated with greater exposure to adverse-selection costs.

5 Data

5.1 FI-2010

The primary starting point will be the publicly available FI-2010 limit order book benchmark dataset introduced by Ntakaris et al. [5]. It contains high-frequency limit order book observations for five stocks traded on the Nasdaq Nordic Helsinki exchange over ten trading days, with information on multiple levels of the bid and ask sides.

FI-2010 provides a practical and computationally manageable starting point for analysing:

- mid-prices,
- bid–ask spreads,
- order-book depth,
- order imbalance, and
- short-term price movements.

5.2 LOBSTER

If access is available through the institute, LOBSTER will be used as a higher-resolution extension. LOBSTER reconstructs NASDAQ TotalView-ITCH limit order book data and provides message-level information including timestamps, event types, order IDs, prices, sizes and order directions [6].

LOBSTER would allow the project to additionally analyse order submissions, executions and cancellations and to study the lifecycle of liquidity provision around shock events.

The project will therefore remain feasible using FI-2010 alone, while LOBSTER will provide a richer extension if access is obtained.

6 Methodology

6.1 1. Liquidity Measures

For each observation, we will calculate the mid-price

$$M_t = \frac{A_t + B_t}{2},$$

where A_t and B_t denote the best ask and best bid.

The bid–ask spread is

$$S_t = A_t - B_t,$$

and the relative spread is

$$RS_t = \frac{A_t - B_t}{M_t}.$$

For K levels of the order book, total depth can be measured as

$$D_t^{(K)} = \sum_{i=1}^{K} V_{i,t}^{bid} + \sum_{i=1}^{K} V_{i,t}^{ask}.$$

Order imbalance will be measured as

$$OI_t = \frac{V_t^{bid} - V_t^{ask}}{V_t^{bid} + V_t^{ask}}.$$

6.2 2. Identification of Liquidity Shocks

Temporary liquidity shocks will be identified using extreme observations of order-flow imbalance and, where appropriate, unusually large short-term price movements or deterioration in liquidity measures.

For example, sell-side and buy-side shocks may initially be defined using the lower and upper tails of the order-imbalance distribution:

$$OI_t < P_5(OI) \qquad \text{or} \qquad OI_t > P_{95}(OI).$$

Robustness checks will use alternative thresholds and shock definitions.

6.3 3. Event-Study Analysis

For each identified shock, market conditions will be examined over an event window surrounding the shock:

$$[-K, +K].$$

The evolution of the following variables will be compared before and after the shock:

- mid-price,
- bid–ask spread,
- order-book depth,
- order imbalance, and
- trading activity.

This will allow us to determine whether the market returns toward its pre-shock liquidity and price conditions.

6.4 4. Dealer-Like Liquidity Provision

Since the available public datasets do not directly identify individual dealers, the project will not claim to observe individual dealer identities.

Instead, we identify *dealer-like liquidity provision* through observable contrarian behaviour. For example, following unusually strong selling pressure, increased buying-side liquidity or buying activity will be interpreted as behaviour consistent with absorption of the selling pressure.

Similarly, increased selling-side liquidity following strong buying pressure will represent the corresponding contrarian response.

6.5 5. Stabilization and Recovery

Price and liquidity recovery will be measured over several post-shock horizons. For example, the change in the mid-price can be studied as

$$\Delta M_{t,k} = M_{t+k} - M_t.$$

Analogous recovery measures will be constructed for spreads and order-book depth.

The key comparison will be between shocks followed by relatively strong liquidity provision and shocks followed by relatively weak liquidity provision.

6.6 6. Adverse Selection

If LOBSTER order-level data are available, the project will examine the subsequent movement of the mid-price after execution of passive limit orders. This provides a measure of the adverse-selection exposure faced by liquidity providers.

7 Statistical Analysis

The primary analysis will use event studies and cross-sectional comparisons between shock events.

As an extension, we will estimate regressions of the form

$$Recovery_i = \beta_0 + \beta_1 LP_i + \beta_2 Volatility_i + \beta_3 Spread_i + \beta_4 Depth_i + \epsilon_i,$$

where LP_i measures the intensity of liquidity provision following shock i.

The coefficient β_1 will indicate whether stronger liquidity provision is statistically associated with faster recovery. The analysis will be interpreted as an association rather than automatically as a causal effect.

8 Expected Outcomes

The project aims to establish whether:

1. temporary liquidity shocks produce measurable deterioration in market liquidity;
2. observable liquidity providers respond contrarianly to extreme order-flow pressure;
3. stronger liquidity provision is associated with faster recovery in prices, spreads and order-book depth; and
4. liquidity provision during severe shocks involves measurable adverse-selection costs.

The project will also compare the empirical findings with the economic mechanisms proposed in the four reference papers.

9 Limitations and Assumptions

The main methodological limitation is that the available datasets do not directly reveal the identity of individual dealers. Therefore, the project uses observable liquidity-provision behaviour as a proxy for the economic role of dealers.

In addition, liquidity shocks are identified from market microstructure variables rather than from an external classification of news events. Consequently, the results will describe the relationship between liquidity-provision behaviour and market recovery rather than establish strict causal effects.

10 References

References

[1] T. S. Ho and H. R. Stoll, "The Dynamics of Dealer Markets under Competition," *The Journal of Finance*, vol. 38, no. 4, pp. 1053–1074, 1983.

[2] Y. Amihud and H. Mendelson, "Dealership Market: Market-Making with Inventory," *Journal of Financial Economics*, vol. 8, no. 1, pp. 31–53, 1980.

[3] B. Biais and P.-O. Weill, "Liquidity Shocks and Order Book Dynamics," NBER Working Paper No. 15009, 2009. doi: 10.3386/w15009.

[4] B. Biais, F. Declerck, and S. Moinas, "Who Supplies Liquidity, How and When?" BIS Working Papers No. 563, 2016.

[5] A. Ntakaris, M. Magris, J. Kanniainen, M. Gabbouj, and A. Iosifidis, "Benchmark Dataset for Mid-Price Forecasting of Limit Order Book Data with Machine Learning Methods," *Journal of Forecasting*, vol. 37, no. 8, pp. 852–866, 2018. doi: 10.1002/for.2543.

[6] LOBSTER, "Limit Order Book System – The Efficient Reconstructor," LOBSTER Data Platform and Documentation. https://lobsterdata.com/